

## **Descriptive analyses**

(what happened?)

According to my analyses and data, this is what happened accordingly.

Eric Mugisha ranks highest in total revenue, followed by Jean Bosco and Aline Uwase.

Running totals and averages show steady growth in customer spending, with spikes in late September.

Quartile segmentation reveals that one-third of customers contribute to the highest revenue bracket.

LEAD/LAG functions show consistent purchasing behavior for some customers and gaps for others.

## **Diagnostic**

(why?)

High revenue concentration suggests a need for targeted retention strategies.

Seasonal spikes may align with promotions or school uniform demand.

Lower quartile customers could benefit from upselling or bundled offers.

Gaps in repeat purchases highlight opportunities for loyalty programs or follow-up campaigns.

## **Insights**

Top Customers: A small group of loyal buyers contributes the largest share of revenue.

Revenue Distribution: Quartile analysis showed that 25% of customers generate more than half of the sales.

Seasonal Trends: Sales are higher around cultural events and holiday seasons.

Growth Monitoring: LAG/LEAD functions help track revenue changes month-to-month, supporting forecasting.

Operational Gaps: Inventory queries highlight products that frequently run out of stock, requiring better supplier coordination.

## **References**

PostgreSQL Documentation: <https://www.postgresql.org/docs/>

Rwanda Development Board (RDB) – Textile Industry Reports

Assignment Guidelines provided by [University/Instructor]

PostgreSQL Window Functions Tutorial –

<https://www.postgresqltutorial.com/postgresql-window-function/>

SQL Shack – Window Functions Overview – <https://www.sqlshack.com/sql-window-functions/>

Mode Analytics – SQL Window Functions – <https://mode.com/sql-tutorial/sql-window-functions/>

Stack Overflow Discussions on RANK & NTILE –  
<https://stackoverflow.com/questions/tagged/window-functions>

Medium Article – Customer Segmentation in SQL –  
<https://medium.com/analytics-vidhya/customer-segmentation-using-sql>

Kaggle Community – SQL Analytics & Ranking Queries –  
<https://www.kaggle.com/learn/advanced-sql>

Rwanda Development Board (RDB) – Textile & Apparel Sector Reports – <https://rdb.rw>

National Institute of Statistics of Rwanda (NISR) – <http://www.statistics.gov.rw>